PA 2 REPORT!!

1. How to execute your program?

在 Visual Studio Code 點擊  執行程式

(1)首先跳出

```
*** Welcome to Positive Binary adder ***
Please enter the 1st number in decimal:
```

並請使用者輸入第一個數

若輸入之值小於-1 或是大於 524287(2^19-1)會顯示錯誤訊息並

請使用者再次輸入值，直到使用者輸入正確的值。

PS:(因為使用 long long int 存取 binary 時只有 19 位數)

```
Please enter the 1st number in decimal: -2
The decimal number you entered is out of range. Please enter the 1st decimal
number again: 545454545
The decimal number you entered is out of range. Please enter the 1st decimal
number again:
```

(2)第一次輸入正確值後，接著輸入第二個值

若輸入之值與前者之和大於 524287 的話會再次跳出錯誤訊息餅

請使用者再次輸入，直到正確為止。

```
Please enter the 1st number in decimal: 5
Please enter the 2nd number in decimal: 524283
The partial sum is out of range. Please enter the 2nd decimal number again:
```

(3)兩個數正確後，會輸出 Decimal 和與 Binary 加法過程與和。

```
*** Welcome to Positive Binary adder ***
Please enter the 1st number in decimal: 5
Please enter the 2nd number in decimal: 6
Calculating... 101 + 110 = 1011
Partial Sum: 11

Please enter the 3rd number in decimal:
```

2.How to compile the program

(1)Deciamaltobinary function 和 Binarytodecimal function :

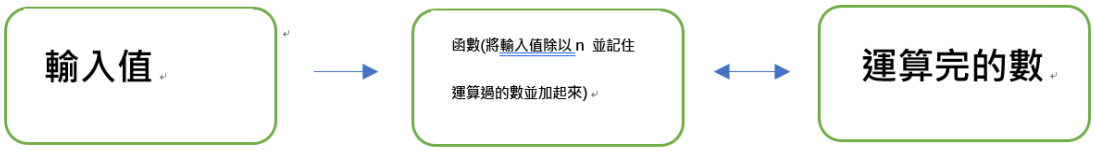
```cpp
#include <iostream>
#include <cmath>

using namespace std;

//function
long long int decimaltobinary(long long int a) { //recursion about decimal to binary
    if (a == 0) {
        return 0;
    }
    else {
        return(a % 2 + 10 * decimaltobinary(a / 2));
    }
}
long long int binarytodecimal(long long int a) { //recursion about binary to decimal
    if (a == 0) {
        return 0;
    }
    else {
        return(a % 10 + 2 * binarytodecimal(a / 10));
    }
}
```

其實兩者大同小異，都運用相似的概念

概念流程圖如下



運用遞迴進行轉換不同形式的數字表示方法

此遞迴的 base case 是 0 ，當輸入函數的值為 0 就會 return 0

並結束這次的函數。

(2) time function

```
23  void timee(long long int g) { //st nd rd th...
24      switch (g % 10) {
25      case 1:
26          switch (g % 100 == 11) {
27          case 1:
28              cout << g << "th";
29              break;
30          default:
31              cout << g << "st";
32          }
33          break;
34      case 2:
35          switch (g % 100 == 12) {
36          case 1:
37              cout << g << "th";
38              break;
39          default:
40              cout << g << "nd";
41          }
42          break;
43      case 3:
44          switch (g % 100 == 13) {
45          case 1:
46              cout << g << "th";
47              break;
48          default:
49              cout << g << "rd";
50          }
51          break;
52      default:
53          cout << g << "th";
54      }
55  }
```

運用 switch case 來判斷輸出 th st nd rd

(3)防呆輸出

就防呆的輸出...懶得打太多次 cout

```
56  void stupidone(long long int g) { //space saving about wrong enter(enter out of range)
57      cout << "The decimal number you entered is out of range. Please enter the ";
58      timee(g);
59      cout << " decimal number again: ";
60  }
61  void stupidtwo(long long int g) { //space saving about wrong enter(sum out of range)
62      cout << "The partial sum is out of range. Please enter the ";
63      timee(g);
64      cout << " decimal number again: ";
65  }
```

(4)二進位加法

```
long long int binaryadder(long long int a, long long int b, long long int c) {
    long long int memory;
    if (a == 0 && b == 0)
        return 0;
    if (a % 10 + b % 10 >= 2) {
        memory = (a % 10 + b % 10 - 2) * c;
        a /= 10;
        b /= 10;
        a += 1;
    }
    else {
        memory = (a % 10 + b % 10) * c;
        a /= 10;
        b /= 10;
    }
    return (memory + binaryadder(a, b, c * 10));
}
```

與 d2b,b2d 有異曲同工之妙，都用相似概念寫遞迴

一位一位相加，當相加大於等於 2 時進位。

(5)main function

```
int main() {
    long long int a, b; //enter
    long long int N = 1; //which time
    int tf; //true false
```

設我所需的參數

```
    while (1) {  //first time enter
        cin >> a;
        if (a == -1) { //quit
            cout << "Bye";
            tf = 0;
            break;
        }
        else if (a < 0 || a >= 524288) { //first time foolproof
            stupidone(N);
            continue;
        }
        else {
            tf = 1;
            break;
        }
    }
```

第一次的輸入(因為第一次和其他次的輸入後的運作方式不太一樣)

```
while (tf) { //N times enter
    cout << "Please enter the ";
    timee(N);
    cout << " number in decimal: ";
    while (cin >> b) {
        if (b == -1) { //quit
            cout << "Bye";
            tf = 0;
            break;
        }
        else if (b < 0 || b >= 524288) { //N times enter foolproof
            stupidone(N);
            continue;
        }
        if (a + b >= 524288) {
            stupidtwo(N);
            continue;
        }
        else break;
    }
    if (b == -1)break;
    cout << "Calculating... " << decimaltobinary(a) << " + " << decimaltobinary(b) << " = " << binaryadder(decimaltobinary(a), decimaltobinary(b), 1)
    N++;
    cout << "Partial Sum: " << binarytodecimal(binaryadder(decimaltobinary(a), decimaltobinary(b), 1)) << endl << endl;
    a += b;
}
```

之後 n 次的輸入

3. Hardness of this assignment how you overcome it

硬要說的話就遞迴吧(想了 10 分鐘?)，因為用 while 一下就寫出來了，不過也因此掌握了基本遞迴的寫法。不過整體來說依舊簡單，只是一些小小限制有些麻煩。

4. Bonus function(s) you implement if any

哈哈哈哈，因為數導太爛而且靠近期中所以只有做一部份

(1)負數二補數轉換

2 補數就是全部倒過來(1->0)(0->1)之後再+1(就是這個 1 在搞 討厭死)，所以會有進位的問題，於是我先寫了進位的 function.

```
long long int carry(long long int a, long long int c) {
    long long int memory;
    if (a == 0)
        return 0;
    if (a % 10 >= 2) {
        memory = (a % 10 - 2) * c;
        a /= 10;
        a += 1;
    }
    else {
        memory = a % 10 * c;
        a /= 10;
    }
    return (memory + carry(a, c * 10));
}
```

接著就重頭戲啦 轉換 2 補數!!! 把正數的二進位表示丟進來處理

(1->0)(0->1)再加 1

```
long long int d2bnegative(long long int x) {

    long long int y = 0;
    long long int z;
    long long int w = 1;
    long long int k;//幾位數
    k = log2(x) + 1;
    while (x > 0) {
        z = x % 2;
        if (z == 1) {
            y += z * w * 0;
            x /= 2;
            w *= 10;
        }
        else {
            y += w;
            x /= 2;
            w *= 10;
        }
    }
    y += 1;;
    y += power(10, k);
    y = carry(y, 1);
    return y;
}
```

(2)負數加減

```
unsigned long long int binaryaddernegative(long long int a, long long int b, long long int c) {
    long long int special;
    if (log10(b) > log10(a)) {
        long long int i = log10(a);
        long long int k = log10(b);
        for (;i < k;i++) {
            special = power(10, (i + 1));
            a += special;
        }
    }
    if (log10(b) < log10(a)) {
        long long int i = log10(b);
        long long int k = log10(a);
        for (;i < k;i++) {
            special = power(10, (i + 1));
            b += special;
        }
    }
    long long int memory;
    if (a == 0 && b == 0)
        return 0;
    if (a % 10 + b % 10 >= 2) {
        memory = (a % 10 + b % 10 - 2) * c;
        a /= 10;
        b /= 10;
        a += 1;
    }
    else {
        memory = (a % 10 + b % 10) * c;
        a /= 10;
        b /= 10;
    }
    return (memory + binaryadder(a, b, c * 10));
}
```

2 補數加減很特殊，位數不夠要補 1，so 我又寫了這些酷酷的東東

然後又因為補完 1 後加法的方式和正數相加一樣，直接引用(懶~

(3) 補數相加可能出現的所有可能性(請詳閱 BONUS code)

簡而言之就爆開啦! 舒服! 爽啊! 刺啊!



5. The completion of the assignment

我覺得應該是全部啦(?

6. Any suggestions about this programming assignment?

我需要一些酷酷的分數

